

Structural monitoring of existing bridges via satellite-based techniques

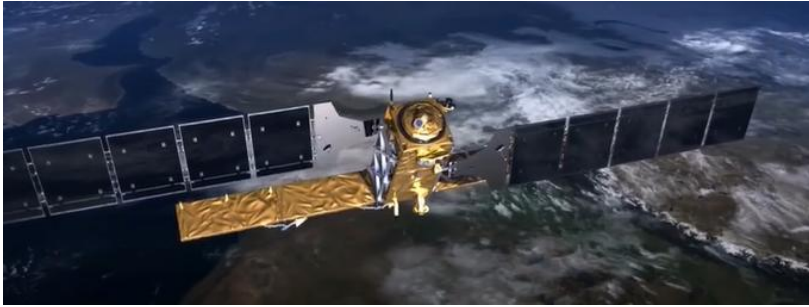
Fabio Di Carlo, Zila Rinaldi, Alberto Meda

Speaker: Fabio DI CARLO
di.carlo@ing.uniroma2.it

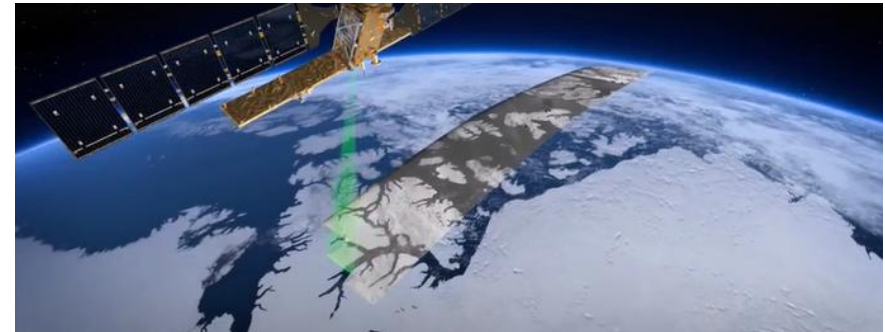
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and Computer Science Engineering*



Structural monitoring via satellite interferometric data



Satellite interferometry is based on the use of a Synthetic Aperture Radar (SAR) system mounted on satellites orbiting the Earth



It enables the monitoring of movements and associated displacement velocity of buildings and infrastructures, with data collected at regular intervals.



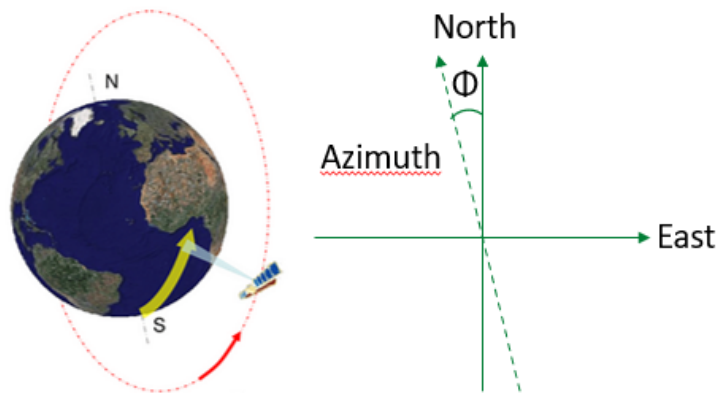
Satellite radar interferometry is an extremely powerful tool for observing the Earth from space, but it is also very delicate to use in practical applications.

Satellite interferometry

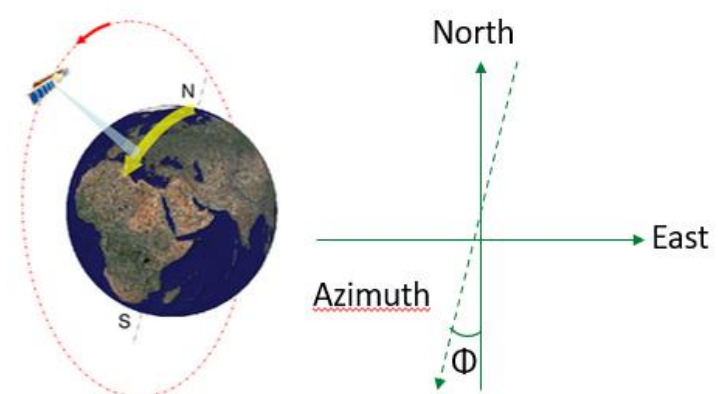
The most commonly used radar satellite systems orbit the Earth in a near-polar, sun-synchronous orbit (such that the satellite passes over the same point on the Earth's surface at the same local solar time) at an altitude of between 500 and 800 km above the Earth's surface

They move along trajectories that are slightly inclined relative to the meridians (at angles of a few degrees), travelling half the orbit from the North Pole to the South Pole (descending orbit) and half from the South Pole to the North Pole (ascending orbit)

Ascending orbit



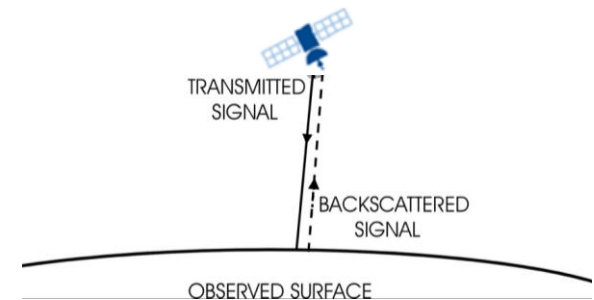
Descending orbit



Satellite interferometry

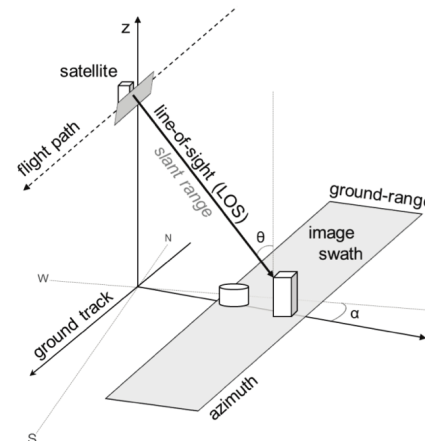
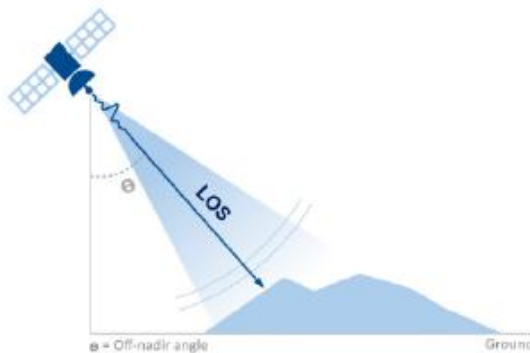
An electromagnetic signal is emitted by a transmitting antenna mounted on the satellite and illuminates the various targets within the observed surface.

The signal is partly absorbed by the illuminated area, partly reflected and partly sent back towards the satellite (backscattered signal), where it is recorded by the receiving antenna.



The phase of the detected signal depends on the distance between the satellite and the ground target, as well as on a range of interfering factors linked to atmospheric conditions and noise.

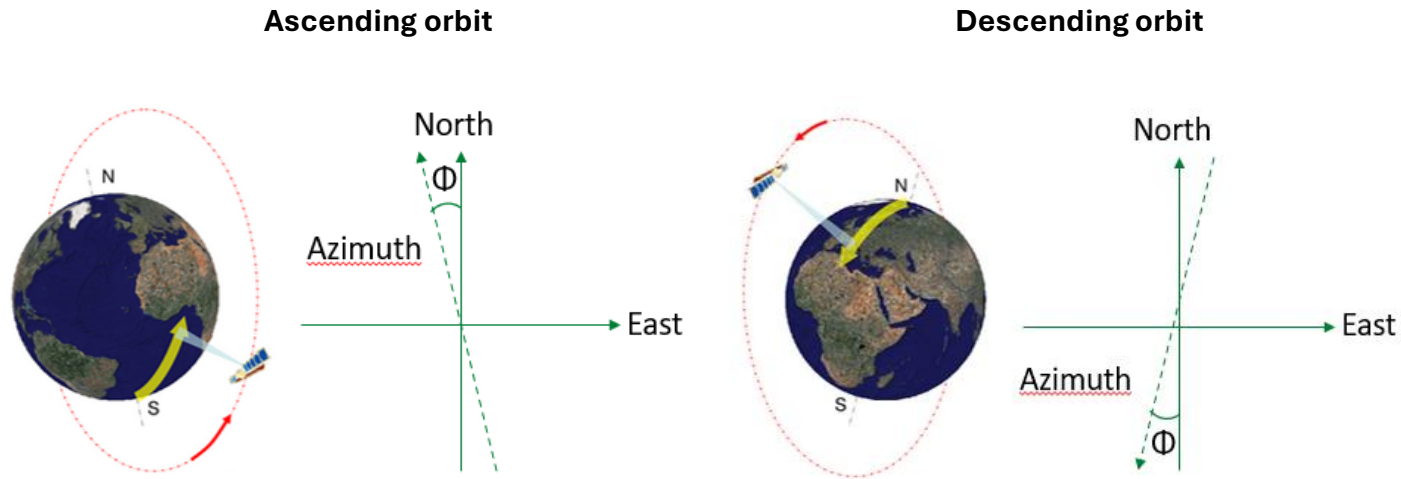
The satellite transmits the signal along a direction known as the Line of Sight, which has an angle of inclination relative to the vertical.



Satellite interferometry

The combination of the satellites' motion along their orbits and the Earth's rotational motion around its axis makes it possible to obtain radar images of the same area at different points in time (revisit time).

Furthermore, the satellite can observe the same ground target when travelling along either an ascending or a descending orbit



The sensors are not affected by other light sources, such as the Sun, and can therefore be used at any time of day or night.

Furthermore, by utilising the microwave frequency range, they can be used in virtually any weather conditions, minimising the impact of clouds or fog, and thus enabling continuous monitoring.

Satellite interferometry

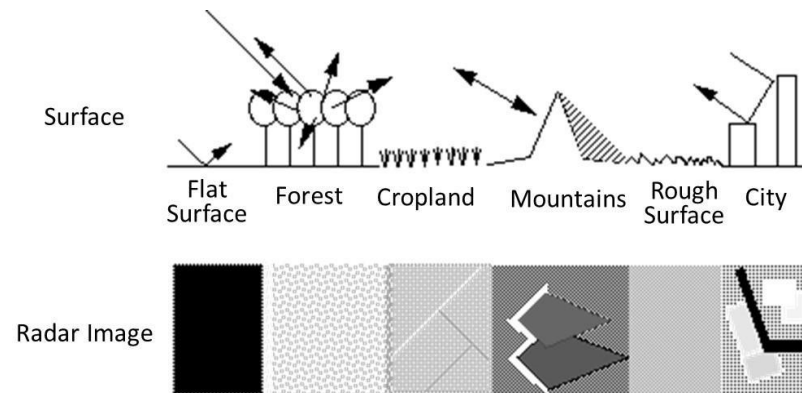
The influence of the physical and geometric characteristics of the illuminated scene

Flat surfaces (e.g. roads and motorways) have very low backscatter

Vegetation is normally quite rough and typically appears uniformly grey

Surfaces sloping towards the sensor (e.g. mountain slopes) exhibit greater backscatter than surfaces sloping away from the radar, and appear very bright in the images.

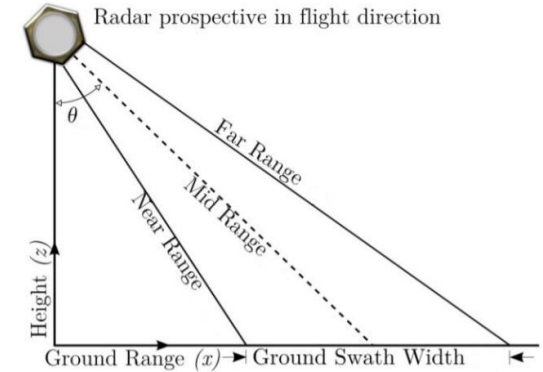
Structures and buildings: the signal emitted by the radar is often reflected several times and returns with little attenuation; the strong backscatter ensures that the image appears very bright



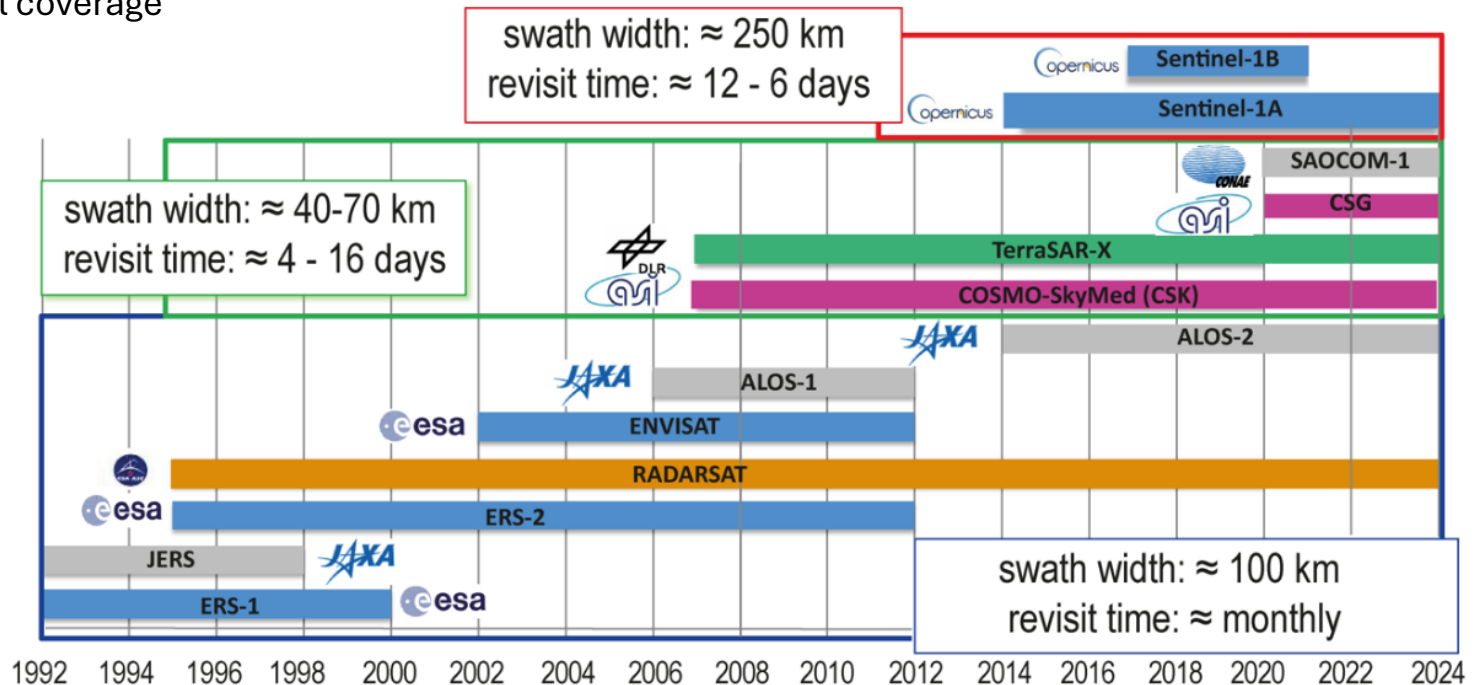
Available satellites and sensors

Over the last thirty years, the field of Earth observation satellites has seen the introduction of numerous sensors with different characteristics in terms of

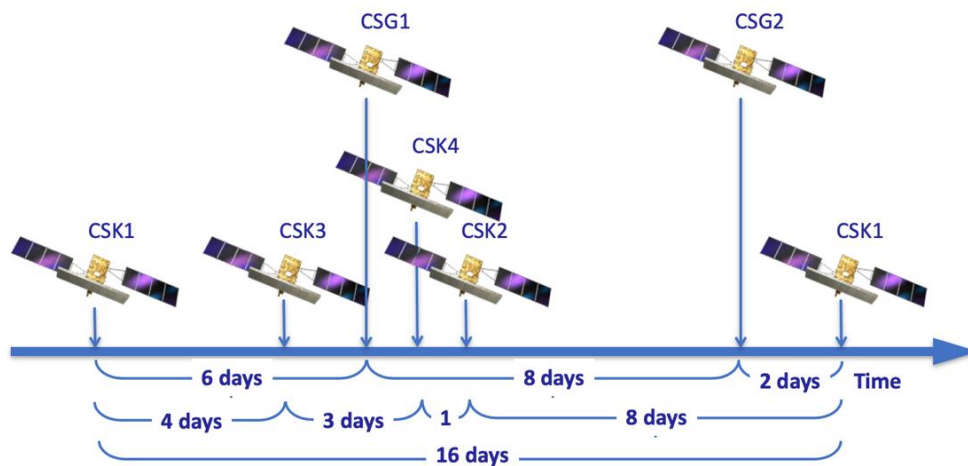
- spatial resolution
- wavelengths
- revisit times
- spatial coverage
- ...



Illuminated area on the ground



COSMO-SkyMed (First and Second Generation CSK/CSG)



CSK-1: 7 June 2007

CSK-2: 9 December 2007

CSK-3: 25 October 2008 – October 2022

CSK-4: 5 November 2010

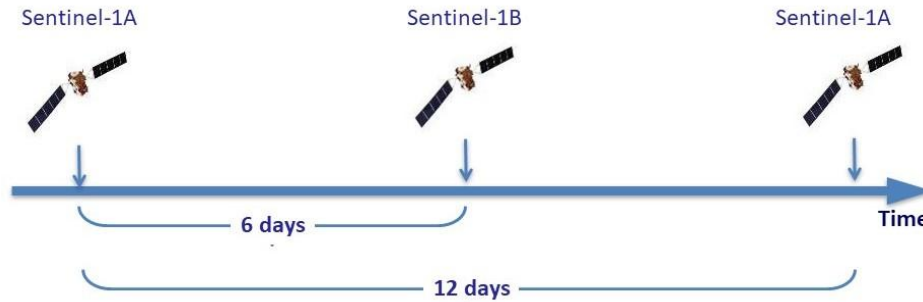
CSG-1: December 2019

CSG-2: January 2022

- SAR sensor, operating in the X-band (wavelength of 3.1 cm)
- Revisit time of 16 days
- Ability to observe the same ground scene on average every four days
- Spatial resolution (StripMap): 3 m × 3 m
- Coverage (StripMap): 40 km



Sentinel-1



Sentinel-1A: 3 April 2014

Sentinel-1B: 25 April 2016 – December 2021 (a failure in the electronic power supply system for the radar instruments)

Open Source data

<https://egms.land.copernicus.eu/>



- SAR sensor, operating in the C-band (wavelength of 5.6 cm)
- Revisit time of 12 days
- Spatial resolution (TOPS): 15 m × 4 m
- Coverage (TOPS): 200 km × 250 km



PROGRAMME OF THE
EUROPEAN UNION



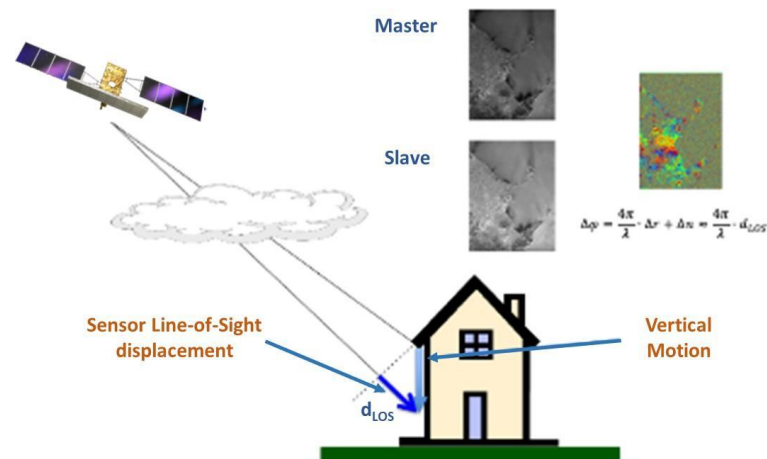
Basic Principles of Differential SAR Interferometry

The DInSAR technique makes it possible to determine the magnitude of the displacement component – in terms of movement towards or away from the sensor – projected along the sensor's line of sight

- Two SAR images are acquired by the satellite over the same area at different times
- One of the two images is chosen as the reference, the 'master' (typically the image acquired first in time), whilst the other is named the 'slave'.
- The variation in the phase of the slave image relative to the master one is evaluated.

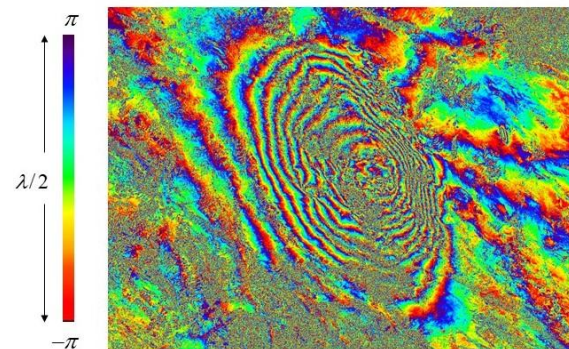
The displacement measurements are differential in both time and space, as they are calculated relative to:

- date of the master image (Reference Date)
- pixel (Reference Point), selected from an area considered stable or with known deformation.

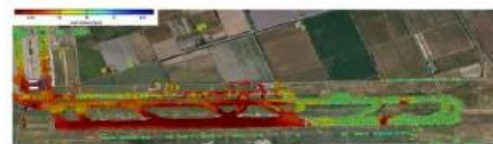


Established fields of application

- Slow-moving landslides
- Effects of the earthquake
- Land subsidence
- ...



A co-seismic interferogram relating to the earthquake that occurred in the city of L'Aquila on 6 April 2009, produced using COSMO-SkyMed SAR images acquired on 4 and 12 April 2009 during ascending orbits.



More questions...



Shift from territorial analyses to studies of individual structures/infrastructures

- The effectiveness of using satellite interferometry in assessing structural behaviour
- What types of movements can be detected?
- What types of structures/infrastructures can be monitored?
- What types of damage can be monitored?
- Identification of the characteristics that sensors should possess to enable the use of data for structural purposes
- Can an accurate diagnosis of a structure be made using satellite interferometry alone?
- Alternatively, what steps would be necessary to arrive at a correct diagnosis?

Experience gained through research projects

- DPC-ReLUIIS 2022-2024 Project



- DPC-ReLUIIS 2024-2026 Project



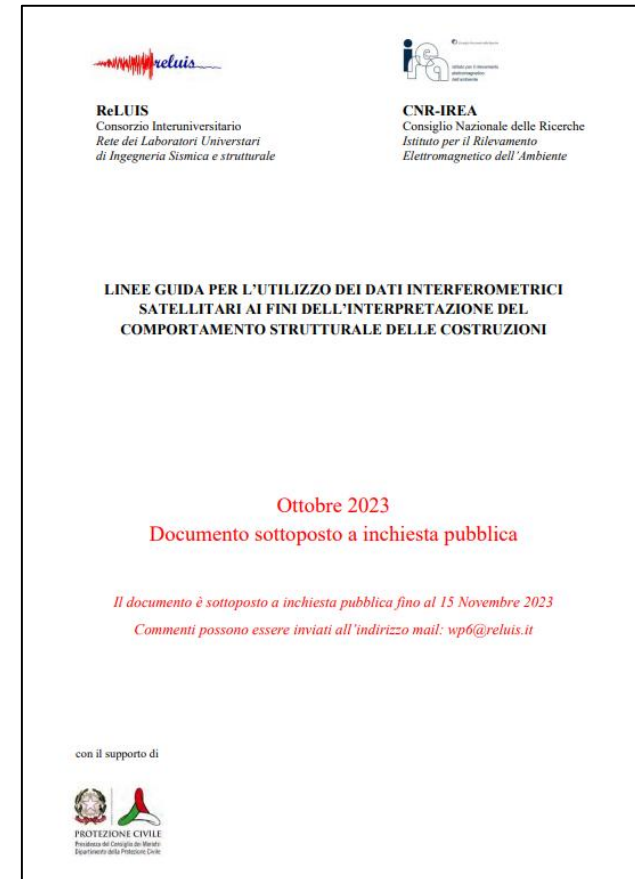
- CSLLPP-ReLUIIS Project



- PRIN 2022 Project – SMUH: Safeguard of Modern Urban Heritage: a cross-disciplinary research for knowledge, monitoring, and risk analysis



Documents



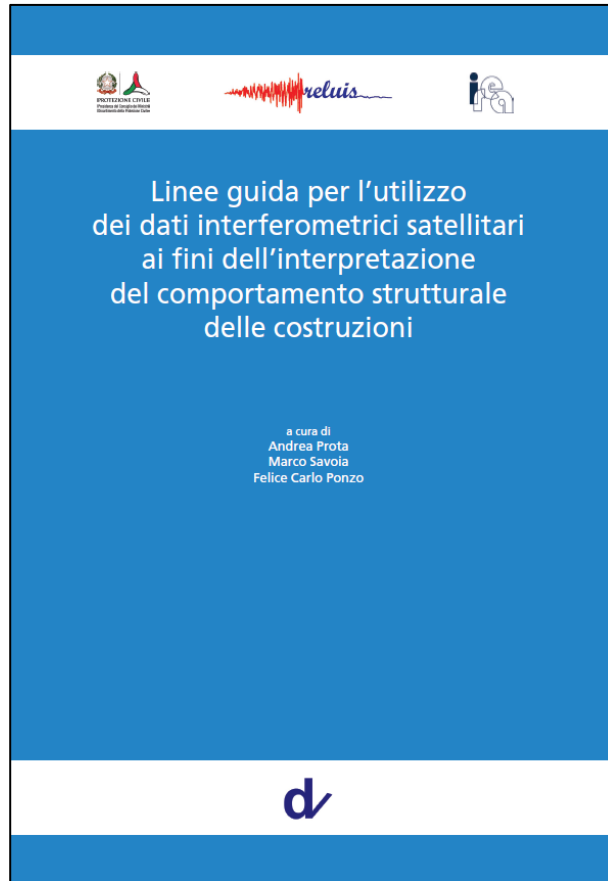
August 2021

October 2023

Documents



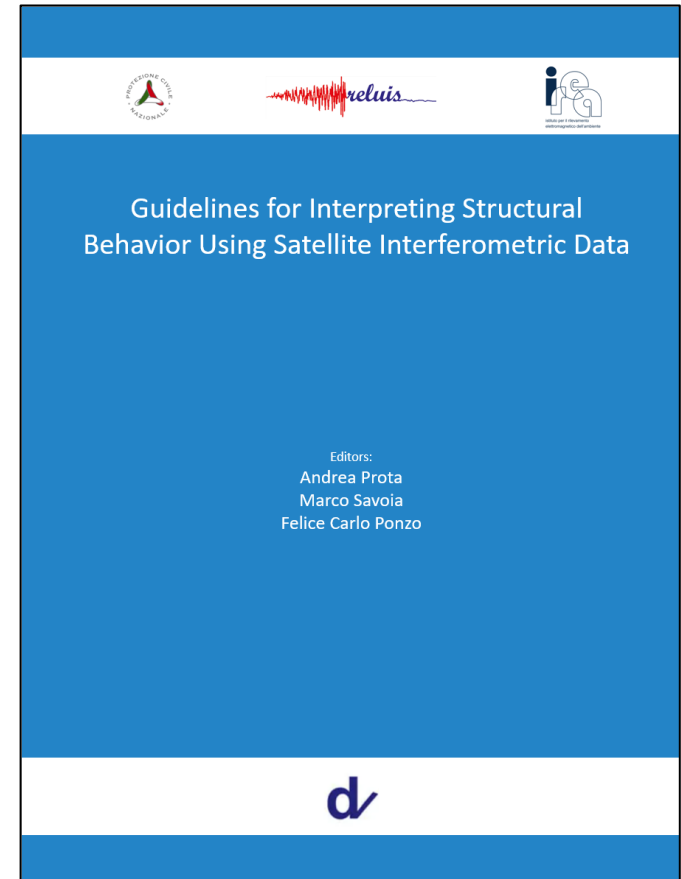
Funded by
the European Union



<https://www.re Luis.it/en/news/4541/guidelines-for-interpreting-structural-behavior-using-satellite-interferometric-data/>



English version



July 2025

February 2026

Measuring points for the ascending and descending datasets

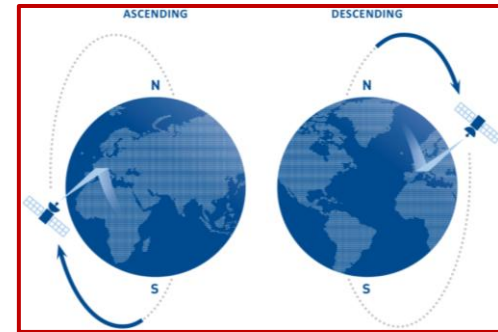
Typically, radar images of an area acquired by the satellite in ascending orbit (ASC) and descending orbit (DES) are processed separately, and two datasets of measuring points are made available for a given time period.

The areas for which the results of analyses based on ascending and descending data are provided can generally be different.



Example of areas where data for ascending orbit (RED) and descending orbit (GREEN) are provided

reference points generally selected in an area considered to be stable



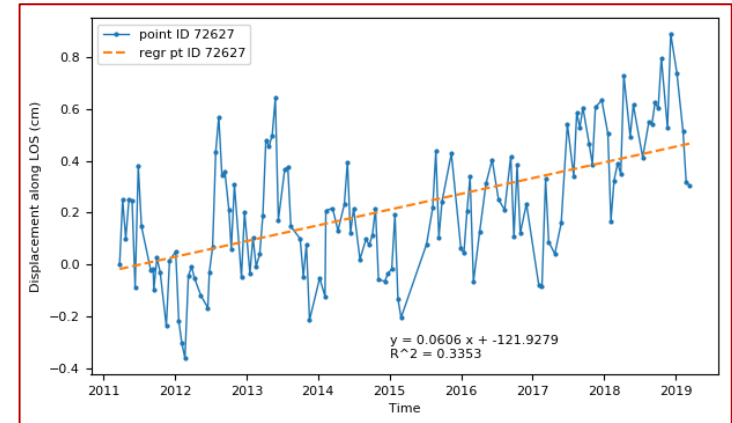
The measuring points obtained from the processing of ascending and descending data do not generally coincide.

- ID
- Coordinates
- Topography
- Average displacement velocity
- Coherence
- LOS directional cosines

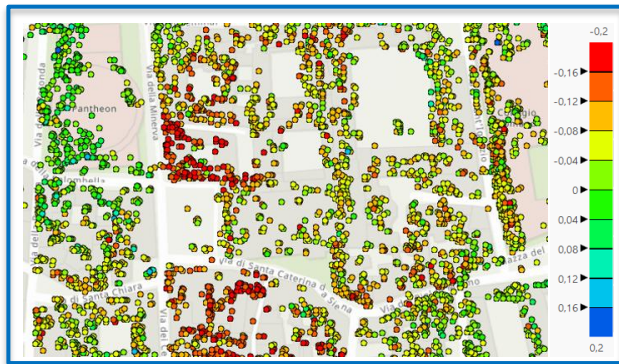
Measuring points for the ascending and descending datasets



For each measurement point, the time history of the displacement component along the LOS is provided.



*Time series of the displacement component along the line of sight (LOS) of the descending orbit (DES) and the **linear regression line***



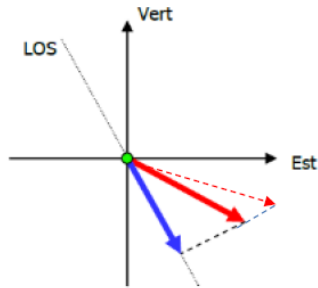
Measurement points coloured according to the average displacement velocity along the LOS

The slope of the linear regression line fitted to the displacement values (red line) gives the **average displacement velocity along the LOS**, at which the measuring point moved during the time interval analysed.

It is possible to create **average displacement velocity maps**. Each measuring point is assigned a colour that indicates the **average displacement velocity along the LOS**.

Interpretation and analysis of data

The knowledge of the displacement component in a single geometry is not sufficient to identify the actual displacement, as several actual displacements may have the same displacement along the LOS

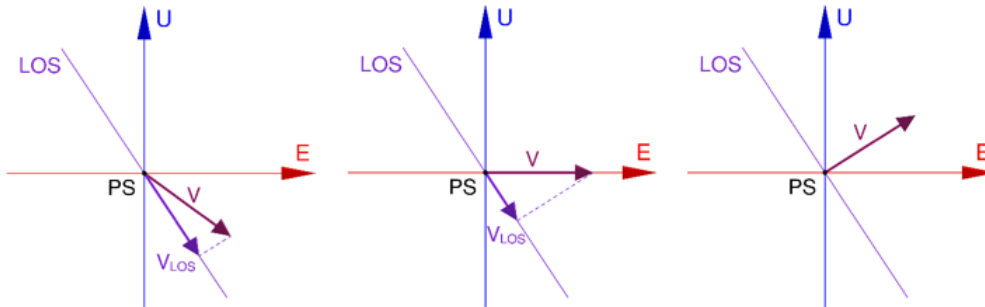


**Approaching displacement
(or velocity) = positive sign**

**Moving away displacement
(or velocity) = negative sign**



The more the direction of the actual displacement vector deviates from the direction of the line of sight, the smaller the component detected by the satellite.



A large displacement along the LOS is certainly an indication that a deformation process is taking place.

On the contrary, a small displacement along the LOS does not guarantee that no deformation is taking place.

Interpretation and analysis of data

USE OF TWO DATASETS – ASCENDING AND DESCENDING

To determine the actual displacement (or displacement velocity), it is necessary to have data for both geometries **at the same point and at the same instant in time**



$$\delta_{LOS,A} = \delta_E \cdot c_{E,A} + \delta_N \cdot c_{N,A} + \delta_U \cdot c_{U,A} \cong \delta_E c_{E,A} + \delta_U c_{U,A}$$

$$\delta_{LOS,D} = \delta_E \cdot c_{E,D} + \delta_N \cdot c_{N,D} + \delta_U \cdot c_{U,D} \cong \delta_E c_{E,D} + \delta_U c_{U,D}$$



Given the displacement (or average displacement velocity) components along the LOS of the two orbits, ASC and DES, at a given point

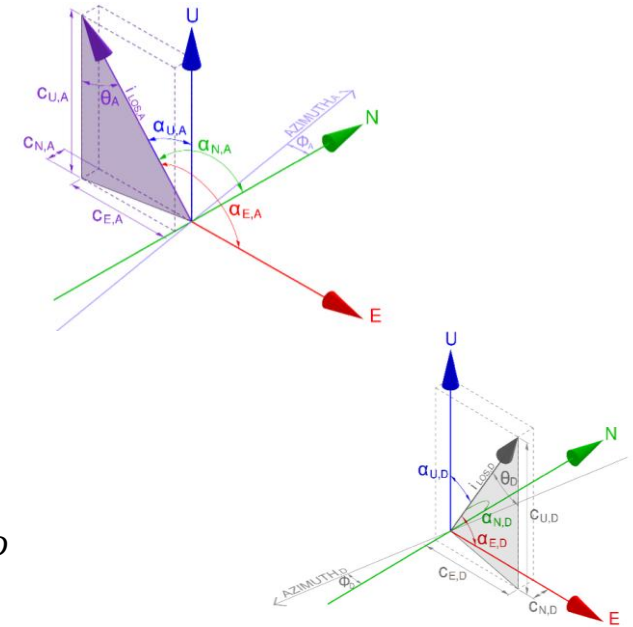
Given the satellite's directional cosines in the two geometries

ASSUMING THAT THE N-S COMPONENT IS ZERO (Due to the inclination of the orbit relative to the Earth's meridians)

it is possible to calculate the vertical and horizontal components (in the E-W direction),
of the displacement / average displacement velocity vector

δ_U

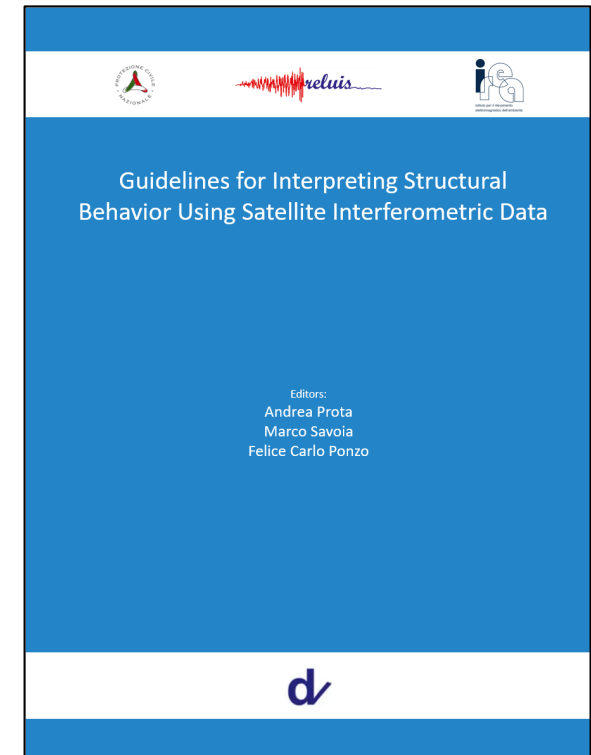
δ_E



Case study § 10.5 Guidelines



“Ansa del Tevere” bridge
also known as “Morandi Bridge at Magliana” or “Morandi
Viaduct”



**Guidelines ReLUIS, CNR-
IREA, Protezione Civile**

Case study description



Designer: Riccardo Morandi

Construction: 1965-1967

Highway: A91 Roma-Fiumicino Motorway

Typology: Cable-stayed

Length: 145 m

Width: 24.20 m



It was the first prestressed concrete cable-stayed bridge built in Rome and the only one, among Morandi's cable-stayed viaducts, that supports a curved road section.

The cable-stayed solution was adopted following a landslide that affected the area where ANAS was building a 640 m long viaduct, part of the state road 201 of the Fiumicino Airport, which was under construction at the time.

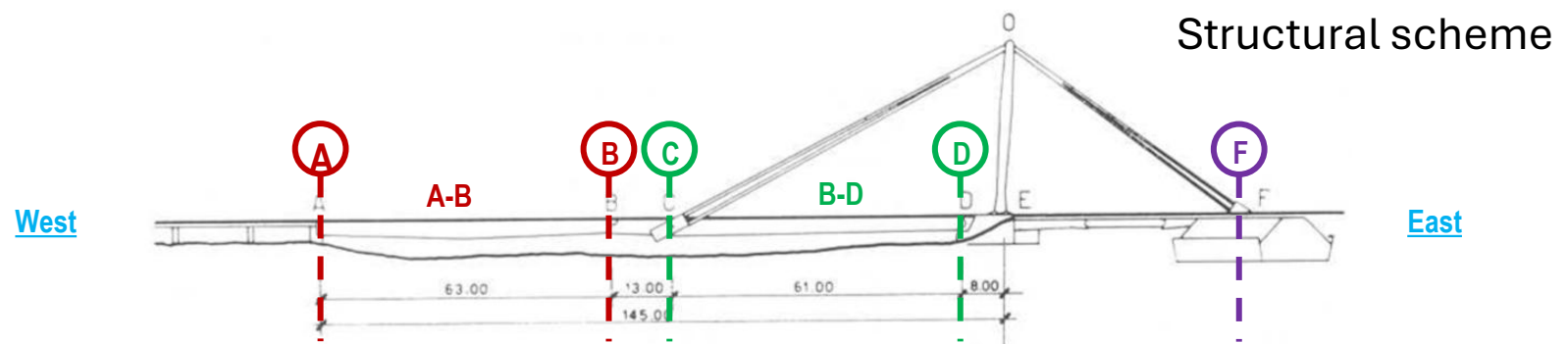
Case study description

Length: 145 m

Width: 24.20 m

A-B: 63 m long, simply supported beam scheme, on the end pier **(A)** of the part of the viaduct that was not affected by the landslide, and on the Gerber saddle **(B)** of the beam **B-D**

B-D: 74 m long, with a hinge at the end section **(D)** and supported in **(C)** by a pair of prestressed concrete tie rods. The west-facing tie rods are connected to the roadway at point **(C)**. The east-facing tie rods, on the other hand, are anchored to the ground at **(F)**.

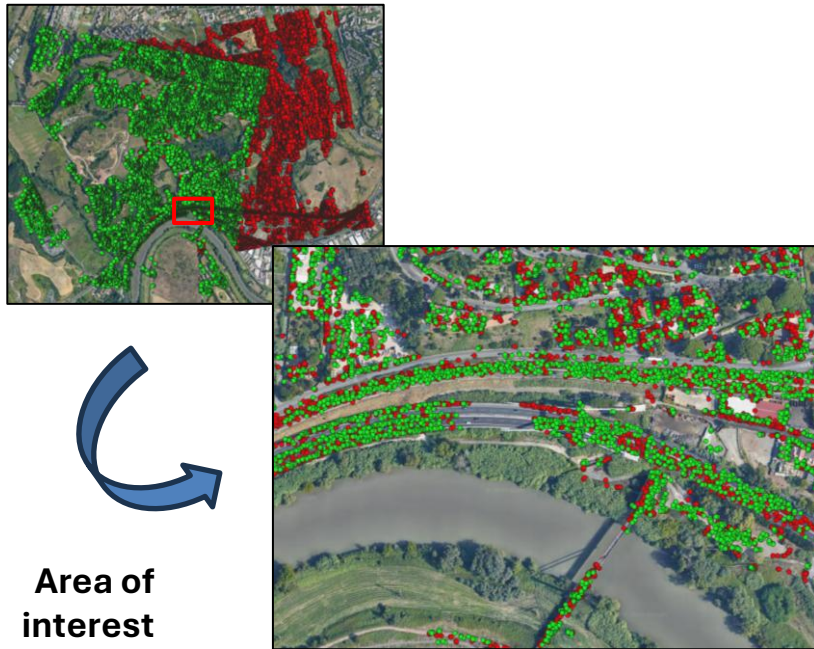


Satellite Data for the Case Study

Data given by CNR-IREA

COSMO-SkyMed satellites
[2011-2019]

Filtering process



Satellite data

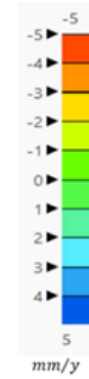
ASCENDING DATASET

DESCENDING DATASET

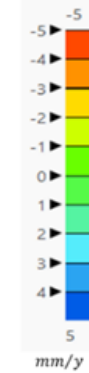
Satellite Data for the Case Study

LOS mean displacement velocity

Ascending dataset



Descending dataset

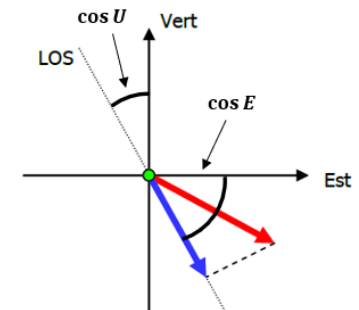


$$\begin{aligned}\delta_{LOS,A} &= \delta_{E-W} \cdot c_{E,A} + \delta_{N-S} \cdot c_{N,A} \\ &+ \delta_{vert} \cdot c_{U,A} \\ &\cong \delta_{E-W} \cdot c_{E,A} + \delta_{vert} \cdot c_{U,A}\end{aligned}$$

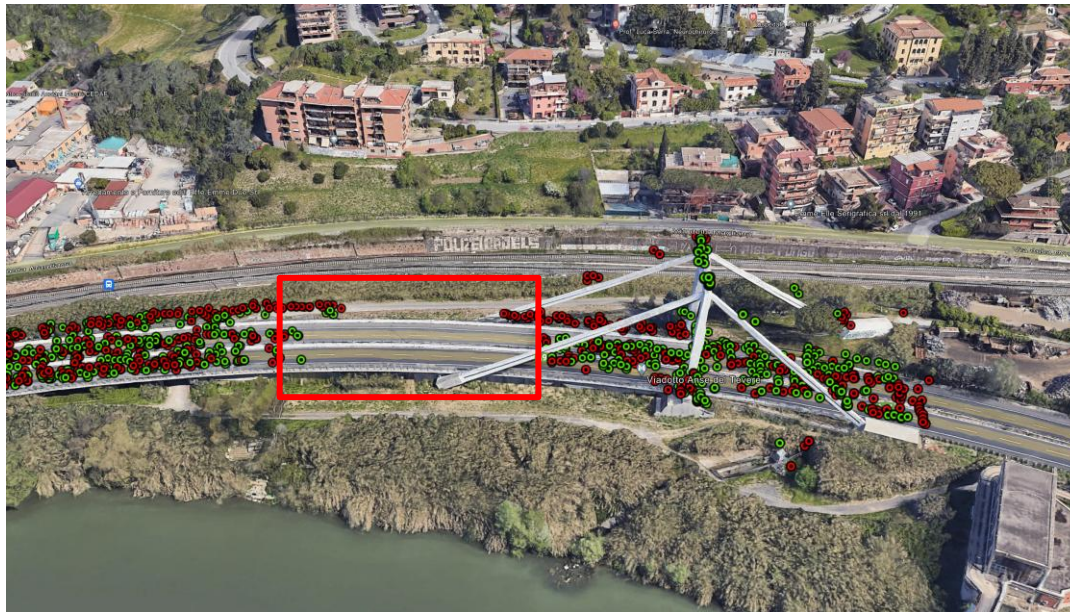
$$\begin{aligned}\delta_{LOS,D} &= \delta_{E-W} \cdot c_{E,D} + \delta_{N-S} \cdot c_{N,D} \\ &+ \delta_{vert} \cdot c_{U,D} \\ &\cong \delta_{E-W} \cdot c_{E,D} + \delta_{vert} \cdot c_{U,D}\end{aligned}$$

δ_{vert} δ_{E-W}

$$\delta_{N-S} = 0$$



Representation and processing of interferometric data



**Presence of an area of the
structure characterised by
the absence of measuring
points (PSs)**

**The reasons behind this
phenomenon are not
always clearly identified**

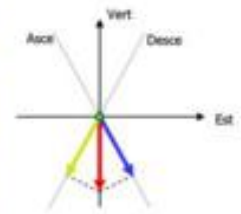
Ascending dataset

Descending dataset

Adoption of different analysis techniques

Different
Analysis
techniques

Nearest Neighbours
Approach

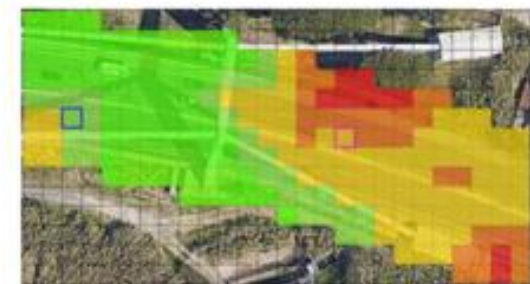


Grid-subsampling
Approach



Vertical displacement velocity

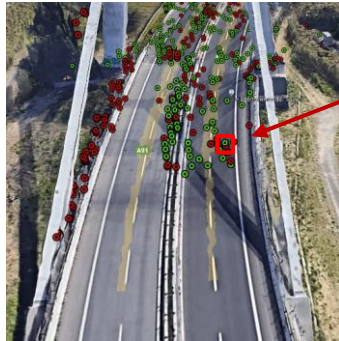
Interpolation and
Combination of Data
Approach



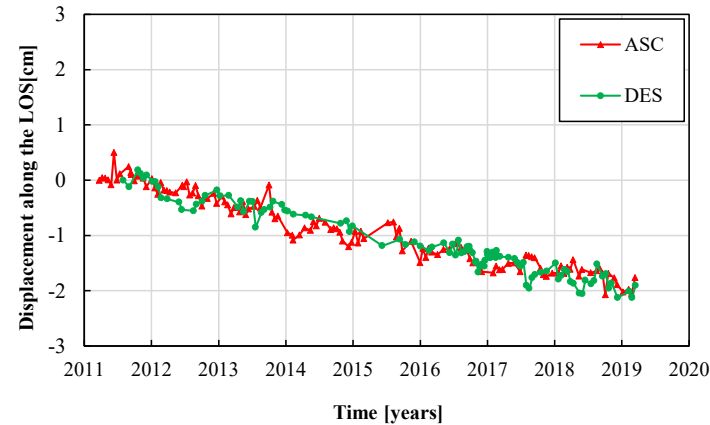
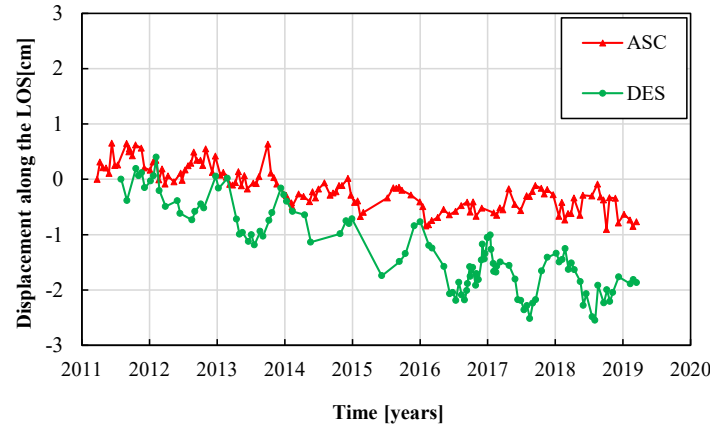
Vertical displacement velocity

Nearest Neighbours Approach

West deck



East deck



Ascending dataset
Descending dataset

Mean velocity

**Evaluation of the
displacement of
the measuring
point**

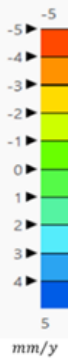
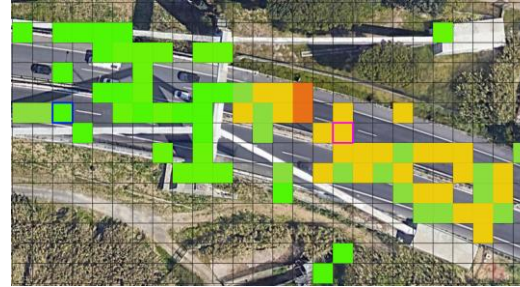
V [cm/year]	West deck	East deck
V_{vert}	-0.246	-0.310
V_{E-W}	-0.141	0.023

Grid-subsampling Approach

Ascending dataset

East-west component

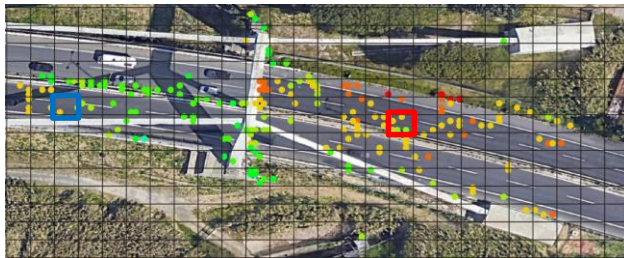
Ascending dataset



Descending dataset

Vertical component

Descending dataset



*Identification of the PSs
falling within each cell*

*6x6m
resampling
cells*

V [cm/year]	West deck	East deck
V_{vert}	-0.187	-0.274
$V_{\text{E-W}}$	-0.131	0.073

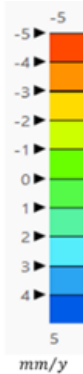
Interpolation and Combination of Data Approach

LOS mean displacement velocity

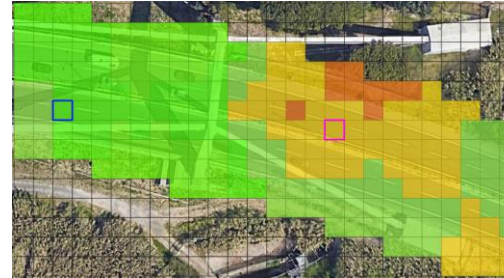
Ascending dataset



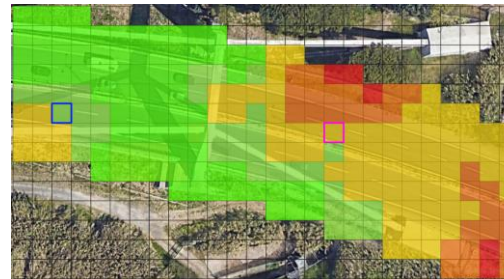
Descending dataset



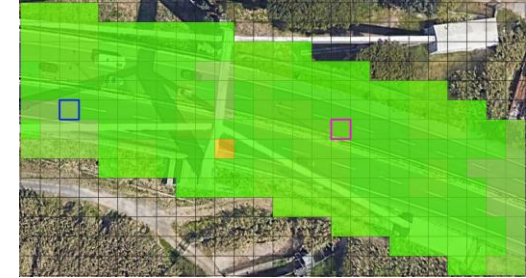
Ascending dataset



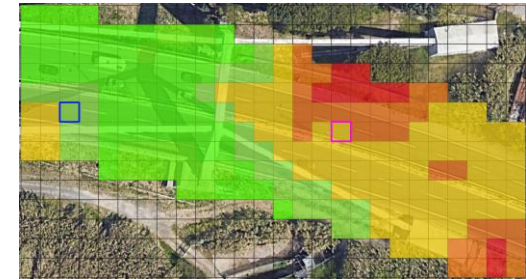
Descending dataset



East-west component



Vertical component



V [cm/year]	West deck	East deck
V_{vert}	-0.164	-0.287
$V_{\text{E-W}}$	-0.115	0.039

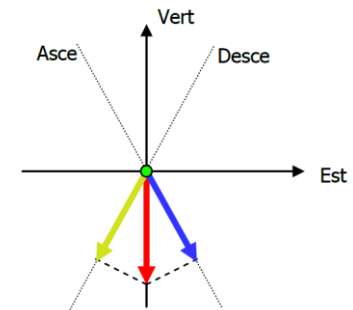
EBK Interpolation technique

All methods show a displacement process, mainly along the vertical direction

	West deck			East deck		
V [cm/year]	Nearest Neighbors (Method 1)	Grid-subsampling (Method 2)	Interpolation and Combination of Data (Method 3)	Nearest Neighbors (Method 1)	Grid-subsampling (Method 2)	Interpolation and Combination of Data (Method 3)
V_{vert}	-0.246	-0.187	-0.164	-0.310	-0.274	-0.287
$V_{\text{E-W}}$	-0.141	-0.131	-0.115	0.023	0.073	0.039



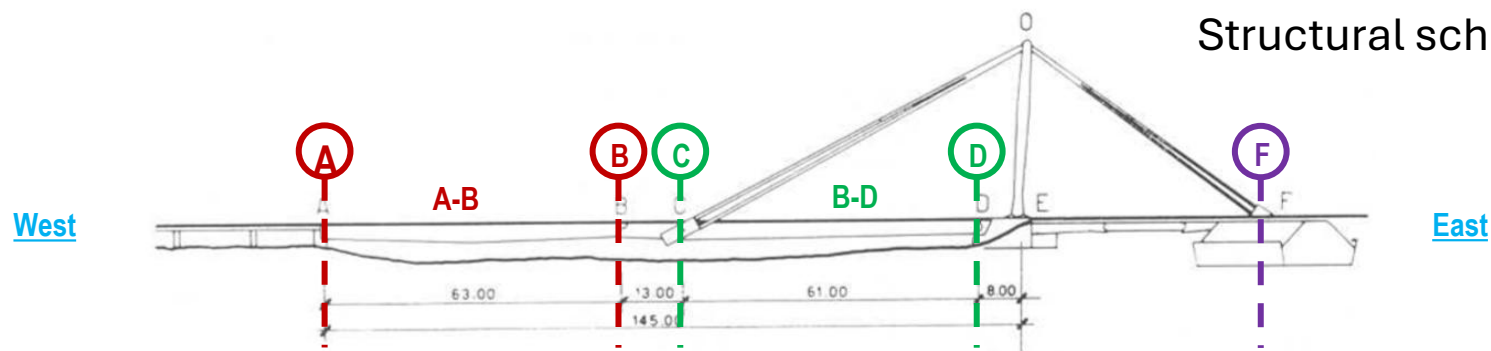
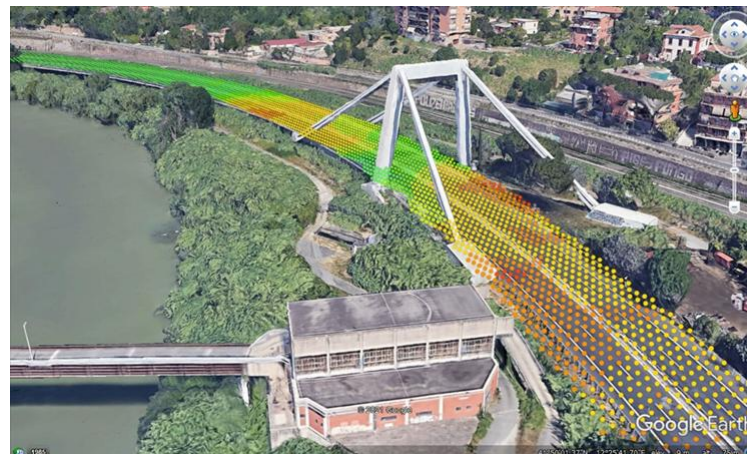
the mean displacement velocity along the LOS exhibits the same sign in both ASC and DES geometries



Furthermore, the displacement phenomenon has a greater intensity in the East deck.

One possible cause could be identified in the existence of a subsidence phenomenon.

Identification of areas for interpolation technique



Structural scheme

Concluding remarks

Some advantages....

- Ability to monitor large areas of territory
- Ability to carry out monitoring without installing fixed ground-based equipment
- Ability to collect data regardless of weather or light conditions
- Long periods of observation
- Ability to use historical time series (at least 10 years)

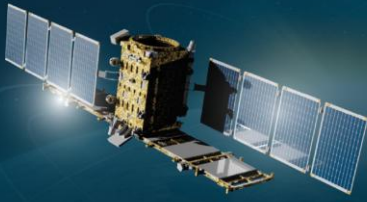
Some difficulties...

- Long time interval between satellite passes (on the order of weeks)
- Effective monitoring of slow deformation phenomena
- Loss of information in case of rapidly moving phenomena
- Accuracy in locating the measurement point (on the order of a few meters)
- Absence of measurement points in areas lacking of reflective objects (e.g. areas covered in vegetation or snow);
- Inability to determine a priori the number and distribution of measurement points
- Inability to obtain measurements in both geometries for the same point at the same time
- Sensor with very low sensitivity to measuring the horizontal N-S component of motion

Future perspectives...

- Use of sensors with different characteristics (wavelength)
- Integration of data provided by constellations operating in different wavelength bands (multi-frequency analysis)
- Integration of satellite data with on-site data
- Commissioning of new satellite constellations, e.g. IRIDE (ESA-ASI) – ‘a constellation of constellations’

Costellazione SAR #1 - Nimbus SAR



PARAMETRO	SPECIFICHE & PRESTAZIONI
ORBITA	MEO 43° e 46° altitudine - 550 km
FREQUENZA RADAR	Banda X
MODI OPERATIVI SAR	Spotlight, Stripmap, ScanSAR
SWATH	2.5 km (Spotlight), 27.5 km (Stripmap), 100 km (ScanSAR)
ANGOLO DI VISTA OFF NADIR	15-50 gradi off Nadir
RISOLUZIONE A TERRA	Da 0.5m a 37m
SAR BANDA TRASMESSA	Fino a 1000 MHz
SAR POLARIZZAZIONE	Pol singola (VV)
POTENZA RF DI PICCO TRASMESSA	3840 W

[Guarda il video](#)

Tutte le caratteristiche preliminari dei satelliti della costellazione SAR #1.

Costellazione SAR #2 - NOX SAR



PARAMETRO	SPECIFICHE & PRESTAZIONI
ORBITA	SSM/LTAN/LTON 18.00 ± 1h
FREQUENZA RADAR	Banda X
MODI OPERATIVI SAR	Spotlight, Stripmap, ScanSAR-2 and ScanSAR-3
SWATH	10 km (Spotlight), 30 km (Stripmap), 80 km (ScanSAR-2), 80 km (ScanSAR-3)
ANGOLO DI VISTA OFF NADIR	15-50 gradi off Nadir
RISOLUZIONE A TERRA	Da 1.5m a 6.0 m
SAR BANDA TRASMESSA	50-300 MHz
SAR POLARIZZAZIONE	Pol singola (VV)
POTENZA RF DI PICCO TRASMESSA	4006 W
MASSA E DIMENSIONI DEL SATELLITE	Nominale 5800kg, 2.2x1.6x1.5m
VITA OPERATIVA	5 anni
MEMORIA DI BORDO	0.5 TB
DATI BATTI TRASMISSIONE A TERRA	Maggiore di 330 Mbps
PROPULSIONE	Chimica
SAR ANTENNA LUNGHEZZA	3,2 m
COMPATIBILITÀ CON LANCIAZIONE	Vega-C, Vega, Ariane 6, PSLV

[Guarda il video](#)

Tutte le caratteristiche preliminari dei satelliti della costellazione SAR #2.



Thanks for your attention

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